

MODULE 11

Wrap-up

& Next

Steps

What we built, mistakes I made, real costs, and
where to go from here

DURATION

10 min

FORMAT

Talking head

TOTAL

~3h35 course

MÓDULO 11 · WRAP-UP

Construa Seu AI COO

O Que Construimos Juntos

Jornada Completa

.

Módulo 1: VPS rodando + OpenClaw instalado + primeiro "oi"

.

Módulo 2: Servidor blindado (UFW, Fail2ban, allowlist)

.

Módulo 3: Agente com personalidade (SOUL, USER, IDENTITY)

.

Módulo 4: Sistema de memória em camadas (daily · topic · MEMORY.md)

.

Módulo 5: Integrações (1Password, Google, YouTube, RapidAPI) + crons

.

Módulo 6: Skills instaladas (superpoderes instantâneos)

.

Módulo 7: Heartbeats + automações proativas

.

Módulo 8: Multi-agentes com leveling system

.

Módulo 9: Sistema imunológico (watchdog, feedback loops, monitoring)

.

Módulo 10: Mission Control (painel visual)

Parabéns!

Você não tem mais um chatbot. Você tem um COO rodando 24/7.

Módulo 11 · Wrap-up

02

MODULE 11 · WRAP-UP

Build Your AI COO

The Mistakes I Made (So You Don't Make Them)

- Token Overflow on Day 2

Context exploded to 173k+ tokens. Agent became slow and expensive. Solution: daily compaction + topic files + max 30 entries.

- Crons That Didn't Execute (3 days)

systemEvent + main = triggers but DOES NOT execute (durationMs ~0ms). Solution: isolated + agentTurn + announce .

- Sub-Agent Stuck in Limbo

I spawned an agent to generate a report. It failed silently. I found out 3 days later.

Solution: always follow-up in 15-30min.

- Security "Open" in the First Days

dmPolicy: open for 2 days. Anyone who found the bot could command it. Solution: allowlist IMMEDIATELY after setup.

- Consolidation Without Extracting Lessons

I compacted memory before extracting lessons. I lost 80% of the value. Solution: ALWAYS extract decisions/lessons BEFORE compacting.

- Systemd Override Overwrote .env

I changed .env but systemd had override with old env. I wasted time debugging. Solution: update BOTH (.env + systemd override).

- Universal Lesson

Every mistake here is avoidable. Follow the course, read the PRDs, test little by little.

Module 11 · Wrap-up

MODULE 11 · WRAP-UP

Build Your AI COO

How Much Does It Cost to Run an AI Agent? (REAL Breakdown)

Infrastructure

ITEM

COST/MONTH

VPS Hostinger (KVM 1)

\$5-10

Domain (optional)

\$1-2/year

Cloudflare Tunnel (optional)

Free

APIs and Models

USAGE

MODEL

COST/MONTH

Direct interaction (~10-15 msgs/day)

Opus

\$15-25

Crons (17/day)

Sonnet

\$5-10

Heartbeats (4x/day)

Haiku

\$0.50-1

RapidAPI (YouTube, Instagram, X)

.

\$0-20

External Tools (Optional)

TOOL

COST/MONTH

1Password (secure credentials)

\$3-5

Supabase (free database)

Free

NocoDB (self-hosted)

Free

Notion (if you choose this option)

\$0-10

Monthly Total

· Without Optimizing

Everything in Opus

Paid APIs unnecessarily

Total: \$60-100/month

· Optimized

Split Opus/Sonnet/Haiku

Free Tools when possible

Total: \$25-45/month

·Perspective

Less than Netflix + Spotify. Cheaper than hiring an intern.

Module 11 · Wrap-up

04

MODULE 11 · WRAP-UP

Build Your AI COO

10 Unbreakable Rules

Learn from the mistakes of those who came before:

dmPolicy = allowlist ALWAYS

Never leave it "open" in production. Anyone can command your agent.

1

Extract lessons BEFORE compressing

If you don't extract before compressing, you lose 80% of the memory value.

2

Crons = isolated + agentTurn + announce

systemEvent in main does not execute. It only triggers and dies (durationMs ~0ms).

3

Model Split: Opus/Sonnet/Haiku

Interaction = Opus, Crons = Sonnet, Heartbeats = Haiku. Saves 70%+.

4

Sub-agents ALWAYS with follow-up

Never "fire and forget". Check in 15-30min or it goes to silent limbo.

5

Credentials in 1Password, NEVER hardcoded

Not even in exposed .env. Use 1Password CLI or environment variables.

6

Backup before structural changes

Config + memory + ROLLBACK.md. If it goes wrong, you go back.

7

Automated weekly security audit

Rate limits and budgets configured

Prevents runaway costs. A bug cannot drain \$500 from the API.

8

Feedback loops for continuous learning

An agent that doesn't remember approve/reject will repeat mistakes forever.

10

Module 11 · Wrap-up

05

MODULE 11 · WRAP-UP

Build Your AI COO

Community and Resources

- OpenClaw Official

Docs: docs.openclaw.ai

GitHub: github.com/openclaw/openclaw

Discord: discord.com/invite/clawd

- ClawHub

Skills marketplace: clawhub.com

Hundreds of ready-to-install skills. Contribute yours!

- References

- Kevin Simback: Leveling system L1-L4

- Bhanu Teja: 10 agents blueprint (3.7M views)

- Eric Siu: "30 OpenClaw Jobs A Day" · immune system

- Reddit Ultimate Guide: [r/ThinkingDeeplyAI](https://www.reddit.com/r/ThinkingDeeplyAI)

- Simon Willison: Security research

The Future of AI Agents

- What's Next

- Agent-to-Agent Communication: agents of different people talking

- MCP (Model Context Protocol): universal skills standard

- Claude Code Integration: agents that actually code

- Multi-Modal: native vision, audio, video

- Edge Deployment: agents running locally (total privacy)

- You are at the forefront

Most people still use ChatGPT as a glorified search engine. You built a COO.

Module 11 · Wrap-up

06

MODULE 11 · WRAP-UP

Build Your AI COO

Next Steps

· Course Feedback

What worked? What was missing? Send me on Discord or Twitter. This course evolves with real feedback.

Last Message

You didn't just build an AI agent. You built an extension of yourself.

Use it wisely. Respect privacy. Contribute to the community.

And remember: the agent is the tool. You are the master.

Welcome to the future. ·

Use Your Agent For 30 Days

Theory is important. Practice is everything. Talk, delegate, test limits.

1

Contribute to the Community

Created a useful skill? Publish it on ClawHub. Found a bug? Report it on GitHub.

Learned something? Share it on Discord.

2

Expand Gradually

Multi-agents, more integrations, custom Mission Control. But slowly-don't break what already works.

3

Document Your Journey

Blog, Twitter, YouTube-tell your story. Others will learn from you.

4

End of Course

Build Your AI COO: From Zero to Personal Agent with OpenClaw

Module 11 · Wrap-up

07